36. Derive a general expression for the equilibrium constant $K_P$ and $K_C$ for the reaction

$$3H_2(g) + N_2(g) \rightleftharpoons 2NH_3(g)$$

37. Write a balanced chemical equation for the equilibrium reaction for which the equilibrium constant is given by expression

$$K_C = \frac{[NH_3]^4 [O_2]^5}{[NO]^4 [H_2O]^6}$$

38. What is the effect of added inert gas on the reaction at equilibrium at constant volume.

39. Derive the relation between $K_P$ and $K_C$.

40. One mole of $PCl_5$ is heated in one litre closed container. If 0.6 mole of chlorine is found at equilibrium, calculate the value of equilibrium constant.

41. For the reaction

$$SrCO_3(s) \rightleftharpoons SrO(s) + CO_2(g),$$

the value of equilibrium constant $K_P = 2.2 \times 10^{-4}$ at 1002 K. Calculate $K_C$ for the reaction.

42. To study the decomposition of hydrogen iodide, a student fills an evacuated 3 litre flask with 0.3 mol of HI gas and allows the reaction to proceed at 500 $^0$C. At equilibrium he found the concentration of HI which is equal to 0.05 M. Calculate $K_C$ and $K_P$ for this reaction.

43. 1 mol of $CH_4$, 1 mole of $CS_2$ and 2 mol of $H_2S$ are 2 mol of $H_2$ are mixed in a 500 ml flask. The equilibrium constant for the reaction $K_C = 4 \times 10^{-2}$ mol$^2$ lit$^{-2}$. In which direction will the reaction proceed to reach equilibrium ?

44. At particular temperature $K_C = 4 \times 10^{-2}$ for the reaction

$$H_2S(g) \rightleftharpoons H_2(g) + \tfrac{1}{2} S_2(g)$$

Calculate $K_C$ for each of the following reaction

i) $2H_2S(g) \rightleftharpoons 2H_2(g) + S_2(g)$

ii) $3H_2S(g) \rightleftharpoons 3H_2(g) + \frac{3}{2} S_2(g)$

45. 28 g of Nitrogen and 6 g of hydrogen were mixed in a 1 litre closed container. At equilibrium 17 g $NH_3$ was produced. Calculate the weight of nitrogen, hydrogen at equilibrium.

46. The equilibrium for the dissociation of $XY_2$ is given as,

$$2XY_2(g) \rightleftharpoons 2XY(g) + Y_2(g)$$

if the degree of dissociation x is so small compared to one. Show that

$2K_P = PX^3$ where P is the total pressure and $K_P$ is the dissociation equilibrium constant of $XY_2$.

47. A sealed container was filled with 1 mol of $A_2(g)$, 1 mol $B_2(g)$ at 800 K and total pressure 1.00 bar. Calculate the amounts of the components in the mixture at equilibrium given that K = 1 for the reaction

$$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$$

48. Deduce the Vant Hoff equation.

49. The equilibrium constant $K_P$ for the reaction



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